TIRA E- COMMERCE DATABASE MANAGEMENT SYSTEM

DATABASE RELATIONSHIP ANALYSIS USING JOINS

1.Combining Multiple Tables

The TIRA database contains different tables that store customer, product, order, payment, and review information.

The tables can be combined using JOIN operations to retrieve related information.

2. INNER JOIN

INNER JOIN returns only the records that have matching values in both tables.

2.1 Customer and Orders

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.First_Name,
  4      c.Last_Name,
  5      o.Order_ID,
  6      o.Order_Date,
  7      o.Total_Amount
  8  FROM Customer c
  9  INNER JOIN Orders o
 10      ON c.Customer_ID = o.Customer_ID;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT
------------
        101 Harini
N                                                         201 23-OCT-26
        2500

        102 Priya
S                                                         202 23-OCT-26
        1500

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME                                          ORDER_ID ORDER_DAT
-------------------------------------------------- ---------- ---------
TOTAL_AMOUNT
------------

        103 Anitha
R                                                         203 23-OCT-26
        3200
```

2.2 Orders and Payment

```
SQL> SELECT
  2      o.Order_ID,
  3      o.Order_Date,
  4      o.Total_Amount,
  5      p.Payment_Mode,
  6      p.Payment_Amount,
  7      p.Payment_Status
  8  FROM Orders o
  9  INNER JOIN Payment p
 10      ON o.Order_ID = p.Order_ID;

  ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_MODE        PAYMENT_AMOUNT
---------- --------- ------------ ------------------ --------------
PAYMENT_STATUS
--------------------
       201 23-OCT-26         2500 UPI                          2200
Success

       202 23-OCT-26         1500 Card                         1200
Success

       203 23-OCT-26         3200 Net Banking                   850
Success
```

2.3 Product and Order_Details

```
SQL> SELECT
  2       p.Product_ID,
  3       p.Product_Name,
  4       od.Order_ID,
  5       od.Quantity,
  6       od.Unit_Price,
  7       od.Subtotal
  8  FROM Product p
  9  INNER JOIN Order_Details od
 10       ON p.Product_ID = od.Product_ID;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
  ORDER_ID   QUANTITY UNIT_PRICE   SUBTOTAL
---------- ---------- ---------- ----------
       102
Moisturizer
       201          1        500        500

       103
Shampoo
       202          2        900       1800

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
  ORDER_ID   QUANTITY UNIT_PRICE   SUBTOTAL
---------- ---------- ---------- ----------

       104
Lipstick
       203          2       1600       3200

       101
Face Wash
```

3. LEFT JOIN

LEFT JOIN returns all records from the left table and matching records from the right table. If there is no match, the right table columns display NULL.

3.1 Display All Customers and Their Orders

```
SQL> SELECT
  2       c.Customer_ID,
  3       c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  4       o.Order_ID,
  5       o.Order_Date,
  6       o.Total_Amount
  7  FROM Customer c
  8  LEFT JOIN Orders o
  9       ON c.Customer_ID = o.Customer_ID;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------
  ORDER_ID ORDER_DAT TOTAL_AMOUNT
---------- --------- ------------
         101
Harini N
        201 23-OCT-26          2500

         102
Priya S
        202 23-OCT-26          1500

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------
  ORDER_ID ORDER_DAT TOTAL_AMOUNT
---------- --------- ------------

         103
Anitha R
        203 23-OCT-26          3200

         107
Arun T
```

4. RIGHT JOIN

RIGHT JOIN returns all records from the right table and matching records from the left table. If there is no match, the left table columns display NULL.

4.1 Display All Orders and Customer Details

```
SQL> SELECT
  2       c.Customer_ID,
  3       c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  4       o.Order_ID,
  5       o.Order_Date,
  6       o.Total_Amount
  7  FROM Customer c
  8  RIGHT JOIN Orders o
  9       ON c.Customer_ID = o.Customer_ID;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------
  ORDER_ID ORDER_DAT TOTAL_AMOUNT
---------- --------- ------------
         101
Harini N
        201 23-OCT-26          2500

         102
Priya S
        202 23-OCT-26          1500

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------
  ORDER_ID ORDER_DAT TOTAL_AMOUNT
---------- --------- ------------


         103
Anitha R
        203 23-OCT-26          3200
```

5. Complete Order Details

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  4      o.Order_ID,
  5      o.Order_Date,
  6      p.Product_ID,
  7      p.Product_Name,
  8      od.Quantity,
  9      od.Unit_Price,
 10      od.Subtotal,
 11      py.Payment_Mode,
 12      py.Payment_Amount,
 13      py.Payment_Status
 14  FROM Customer c
 15  INNER JOIN Orders o
 16      ON c.Customer_ID = o.Customer_ID
 17  INNER JOIN Order_Details od
 18      ON o.Order_ID = od.Order_ID
 19  INNER JOIN Product p
 20      ON od.Product_ID = p.Product_ID
 21  INNER JOIN Payment py
 22      ON o.Order_ID = py.Order_ID
 23  ORDER BY o.Order_Date DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID ORDER_DAT PRODUCT_ID
---------- --------- ----------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY UNIT_PRICE   SUBTOTAL PAYMENT_MODE         PAYMENT_AMOUNT
---------- ---------- ---------- -------------------- --------------
PAYMENT_STATUS
--------------------
        101
```

6. Customer Purchase History

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  4      o.Order_ID,
  5      o.Order_Date,
  6      p.Product_Name,
  7      od.Quantity,
  8      od.Subtotal
  9  FROM Customer c
 10  INNER JOIN Orders o
 11      ON c.Customer_ID = o.Customer_ID
 12  INNER JOIN Order_Details od
 13      ON o.Order_ID = od.Order_ID
 14  INNER JOIN Product p
 15      ON od.Product_ID = p.Product_ID
 16  ORDER BY c.Customer_ID, o.Order_Date;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------
  ORDER_ID ORDER_DAT
---------- ---------
PRODUCT_NAME
--------------------------------------------------------------
  QUANTITY   SUBTOTAL
---------- ----------
        101
Harini N
       201 23-OCT-26

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------
  ORDER_ID ORDER_DAT
---------- ---------
PRODUCT_NAME
--------------------------------------------------------------
  QUANTITY   SUBTOTAL
```

7. Customer Purchase History with Payment

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  4      o.Order_ID,
  5      o.Order_Date,
  6      p.Product_Name,
  7      od.Quantity,
  8      od.Subtotal,
  9      py.Payment_Mode,
 10      py.Payment_Status
 11  FROM Customer c
 12  INNER JOIN Orders o
 13      ON c.Customer_ID = o.Customer_ID
 14  INNER JOIN Order_Details od
 15      ON o.Order_ID = od.Order_ID
 16  INNER JOIN Product p
 17      ON od.Product_ID = p.Product_ID
 18  INNER JOIN Payment py
 19      ON o.Order_ID = py.Order_ID
 20  ORDER BY c.Customer_ID;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID ORDER_DAT
---------- ---------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY   SUBTOTAL PAYMENT_MODE          PAYMENT_STATUS
---------- ---------- -------------------- --------------------
         101
Harini N
       201 23-OCT-26

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID ORDER_DAT
```

8. Multi-Table Business Report

```
SQL> SELECT
  2      c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  3      COUNT(DISTINCT o.Order_ID) AS Total_Orders,
  4      SUM(od.Subtotal) AS Total_Purchase,
  5      SUM(od.Quantity) AS Total_Products
  6  FROM Customer c
  7  INNER JOIN Orders o
  8      ON c.Customer_ID = o.Customer_ID
  9  INNER JOIN Order_Details od
 10      ON o.Order_ID = od.Order_ID
 11  GROUP BY c.First_Name || ' ' || c.Last_Name
 12  ORDER BY Total_Purchase DESC;

CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_ORDERS TOTAL_PURCHASE TOTAL_PRODUCTS
------------ -------------- --------------
Harini N
           1           3500              4

Anitha R
           1           3200              2

Priya S
           1           1800              2
```

9. Payment Transaction Report

```
SQL> SELECT
  2      c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  3      o.Order_ID,
  4      o.Order_Date,
  5      o.Total_Amount,
  6      py.Payment_Mode,
  7      py.Payment_Amount,
  8      py.Payment_Status
  9  FROM Customer c
 10  INNER JOIN Orders o
 11      ON c.Customer_ID = o.Customer_ID
 12  INNER JOIN Payment py
 13      ON o.Order_ID = py.Order_ID
 14  ORDER BY o.Order_Date DESC;

CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_MODE         PAYMENT_AMOUNT
---------- --------- ------------ -------------------- --------------
PAYMENT_STATUS
--------------------
Harini N
       201 23-OCT-26         2500 UPI                          2200
Success

Anitha R
       203 23-OCT-26         3200 Net Banking                   850
Success

CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_MODE         PAYMENT_AMOUNT
---------- --------- ------------ -------------------- --------------
PAYMENT_STATUS
--------------------

Priya S
       202 23-OCT-26         1500 Card                         1200
Success
```